

Daniel Ejakov

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Data science student with a strong theoretical background with practical hands-on experience in data science, programming, data collection, and field work.

EDUCATION

Honors College at Michigan State University, East Lansing, MI, 2023-2025

B.S. Computational Data Science, Economics minor, 3.6 GPA

WORK EXPERIENCE

Mathematical Biology Tick Research Lab — Old Dominion University (Norfolk, VA), Summers 2024 and 2025

2025

- Developed an end-to-end video analysis pipeline to quantify movement patterns of tick species that are understudied (Python, Linux)
- Designed and implemented standardized protocols for specimen collection, maintenance, recording, and tracking
- Collected specimens across multiple field sites in several states; managed transport and metadata recording
- Processed samples in a multi-year database and developed mathematical models to analyze movement behavior.
- Applied data-processing and statistical techniques (Python, R, Unix, PCA, Seaborn, Matplotlib) to extract interpretable insights
- Supervised a small team to ensure accurate data collection, troubleshoot protocol issues, and iterate experimental design

2024

- Conducted and coordinated extensive field collections across Virginia, Maryland, Delaware, and North Carolina
- Performed meticulous data entry and sample processing in a lab environment, developing strong attention to detail and data-quality discipline
- Tested and evaluated *TickBot*, a robotic device designed for tick control; collected performance data, analyzed effectiveness across environments, and documented design limitations and potential improvements

Soccer Referee, United States Soccer Federation Certified, MI, 2018-2022

- Officiated 100+ competitive matches, applying conflict-resolution and game-management skills under pressure.
- Enforced rules consistently, communicated decisions clearly to players and coaches, and maintained control in fast-paced, high-stakes situations.

To-Go Expediter — Chili's, 2021–2022

- Coordinated high-volume to-go operations and assumed leadership roles during shortages.
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PROJECTS

Video Analysis Tick Population Differentiation, Old Dominion University, 2025

- Developed pipeline to analyze behavioral differences across tick populations (Python, Unix, R); research papers pending.

NBA MVP Predictor — Python 2024

- Developed model to predict NBA MVP winners using NBA dataset with every player since 1979
- Random Forest model with SMOTE (Synthetic Minority Over-sampling Technique) and season-loop validation

Flask User Authentication Backend — Python, SQL, 2025

- Developed user authentication to understand API security
 - Implemented JWT generation and verification, secure hashing and role-based access
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TECHNICAL SKILLS

- Python, R, Unix/Linux, SQL, C, C++, scikit-learn, Matplotlib & Seaborn, Pandas, Git/GitHub, Flask
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RELEVANT COURSEWORK

Algorithms: Algorithm Engineering, Algorithms and Data structures

Data Science / ML: Machine Learning; Fundamentals of Data Science, Bioinformatics, Computational Modelling

Systems / Cloud: Information Management & the Cloud